Resilient EMG Classification to Enable Reliable Upper-Limb Movement Intent Detection

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***Abstract*— Reliable control of assistive devices using surface electromyography (sEMG) remains an unsolved task due to the signal's stochastic behavior that prevents robust pattern recognition for real-time control. Non-representative samples lead to inherent class overlaps that generate classification ripples for which the most common alternatives rely on post-processing and sample discard methods that insert additional delays and often do not offer substantial improvements. In this paper, a resilient classification pipeline based on Extreme Learning Machines (ELM) was used to classify 17 different upper-limb movements through sEMG signals from a total of 99 trials derived from three different databases. The method was compared to a baseline ELM and a sample discarding (DISC) method and proved to generate more stable and consistent classifications. The average accuracy boost of $\approx$ 10% in all databases lead to average weighted accuracy rates higher as 53,4% for amputees and 89,0% for non-amputee volunteers. The results match or outperform related works even without sample discards.**

***Index Terms*— Upper-limb, extreme learning machines, EMG, reliability, resilient classification, robust classification.**

I. Introduction

DIVERSE studies have been explored sEMG as a control signal for natural interfaces [1]–[4]. Although its potential to enable urging human necessities [5]–[7], the technology remains incipient, and regarding robust classification, much is yet to be explored [8]–[11]. The main limitations for robust control are related to the sEMG stochastic behavior, affected by factors such as force intention, biomechanics [9], electrode positioning and long-term use [12]. Lately, diverse signal filtering [13]–[15] and classification strategies have been explored to compensate constraints imposed by sEMG nature and its experimental restrictions. Recent approaches have been focusing on robust strategies to compensate classification errors in movement transitions [16]–[18], different limb positions [10], and optimal feature selection [19]–[21].

The classifier efficiency is usually related to the accuracy rate obtained while specific tasks are executed. Typically, models formed from sEMG signals suffer from latency at movement transitions and classification ripples due to class overlap that limits proprioceptive control. Since pre-processing methods and traditional classifiers are effective just to some extent, additional layers involving Deep-Learning and post-processing have been incorporated into models aspiring to provide more classification robustness [7], [22].

Robust sEMG classification aims to prevent classification ripples and provide more consistent and stable outputs. Despite the proportionality between accuracy and signal length, a simplistic solution based on longer sEMG segments also precludes optimal control by inserting delays to the process [23]–[25]. Thus, using sEMG for robust control requires more sophisticated approaches, while the signal segmentation must balance accuracy and responsiveness for proprioceptive control. As the pre-processing methods and standard classifiers alone are not capable of offering an optimal sEMG control, state of the art methods have been mainly focusing on post-processing and traditional Major-Voting (MV) approaches.

Using a hybrid MV technique, Wahid *et al.* [25] classified six hand movements to compensate for the accuracy decay as a consequence of shorter segmentation. The state of the art usually focuses on more complex approaches and classification "confidence scores". Based on a confidence metric, the non-reliable classifications are either ignored [26] or reprocessed to redefine the output label [10], [27]. Using the confidence score approach, Scheme *et al.* [26] developed a method to define a threshold with Fitt's Law to ignore samples with insufficient confidence. Similarly, Amsuss *et al.* [27] used the maximum likelihood and the sEMG envelope as input for an auxiliary ANN to provide more consistent outputs. In their work, Betthauser *et al.* [10] defined a confidence metric to regulate the use of sparse signal representation and increase robustness for different limb positions. A Reliable ELM was also designed by Cene and Balbinot *et al.* [17] using a self-generated metric to ignore sample outliers and enhance the classification for different databases, including amputees. In their works, Robertson *et al.* [8], [11] have also developed methods of sample discard based on different threshold parameters. Their studies concluded that a significant reduction of systemic errors depends on the threshold level and that an excessive threshold level causes over-rejection (including the correct classes) that impairs the system usability. In general, alternatives involving reliability metrics are still incipient, and much is yet to be explored involving system integration and parameter definition.

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Although the capacity of the methods based on sample discard to increase accuracy, they also affect proprioceptive control by forcing the system to maintain the last reliable classification. As alternative, Cene *et al.* [28] proposed the Smoothed *argmax* ELM (SMT), where ripples are mitigated by *argmax* adaptive filtering before the label attribution. Therefore, the method is capable of tackling shortcomings from MV and confidence score techniques. The MV methods are often coarser approaches since they act only at the system output level, which is frequently insufficient to increase accuracy and inserts delay due to sample buffering. Thus, clear advantages in favor of SMT are that the method is agnostic to confidence scores and it does not discard any samples nor requires any parameters for thresholds. Although its potential for accuracy rate boost, remaining errors were still observed at movement transition where delays depict the classifier's response.

In this paper, we explore a resilient pipeline for sEMG classification to perform a reliable upper-limb movement classification avoiding data discard. Our methods rely on the identification of active movements and reconsideration of *argmax* values used in the classification process. This helped to mitigate errors in movement transition by compensating the lack of representativity for samples in this region and maximize the system sensibility and accuracy. Three databases formed of 99 trials were used to evaluate the resilient classifier that even without sample discard, proved to achieve higher accuracy compared to more traditional alternatives described in the literature and the standard ELM classifier (STD).

II. Materials and Methods

A. sEMG Databases

The methods were used to process datasets from the Exercise B of NINAPro database 2 (11 amputees), NINAPro database 3 (40 non-amputees) [29], and the IEE database (4 non-amputees) [30]. The data from amputees was titled "AMP" while the non-amputees data was denominated "NAMP". Each subject from the IEE database performed tree repetitions of four different assays (IEE A, IEE B, IEE C, and IEE D) forming a total of 48 trials. The block diagram representing the workflow of our system and the different classification methods is presented in Figure 1.

All databases use the same electrode positioning: 12 sEMG channels with channels 1-8 placed around forearm approximately at 3cm of the radio-humeral joint and channels 9-12 placed at fingers extensor and flexor, biceps and triceps, respectively. The IEE database is composed of four different trials (IEE-A to IEE-D) where the subjects perform respectively six and ten repetitions of each movement sequentially in IEE-A and IEE-B and the same number of repetitions randomly for IEE-C and IEE-D. For the NAMP and AMP databases, the six repetitions of each movement were performed sequentially.

The movements performed can be divided into three major groups: 1) Finger Movements - *Thumb up; Extension of middle and index finger and flexion of other fingers; Flexion of little and ring fingers and extension of others; Thumb flexion; Abduction of all fingers; Hand closure; Pointing index and Abduction of extended fingers*. 2) Torsion movements - *Wrist supination (axis: middle finger); Wrist pronation (axis: middle finger); Wrist supination (axis: little finger) and Wrist pronation (axis: little finger)*. 3) Wrist-related movements: *Wrist flexion; Wrist extension; Wrist radial deviation; Wrist ulnar deviation and Wrist extension with the hand closed*. A rest time of 3s for databases AMP and NAMP, 7s for the sequential IEE databases, and 9s for the random IEE databases was used between each movement repetition. Thus, the rest is the predominant class in all databases (38% for NINAPro, 58% for IEE-A and IEE-B, and 64% for IEE-C and IEE-D).

B. Pre-Processing and Feature Extraction

All the signal processing and classification were performed individually for each subject. The sEMG was rectified and filtered using our Anthonyan Vardan Transform (AVT) filter [17], [30] and thereafter segmented and normalized by channel. The signal segmentation considered configurations of 200+10ms, 300+10ms, and 400+10ms based on [24], [29], [31] to determine the most efficient configuration related to the accuracy rate. Moreover, a 10ms of increment was used to favor the proprioceptive control and its time constraints [23], [24]. Root Mean Square, Variance, Mean Amplitude Value, and Standard Deviation were used along with Differential Enhanced Signal [17] for signal representation. The first two-thirds of repetitions were used to train and the final third to test the classifier.

C. Extreme Learning Machines

The ELM classifier is defined using a Moore-Penrose pseudoinverse in its standard version, as presented in Equation (1), and methods such as Ridge-Regression on the regularized version of the classifier [32], [33]. The system is solved when a pseudoinverse matrix $\mathbf{H}^{\dagger}$ is used to find the model β according with the input data in $\mathbf{H}$ considering the labels $\mathbf{T}$. Based on previous works that favored the regularized form [30], the Ridge-Regression model was used in this paper. The regularized pseudoinverse with C as the regularization factor is defined as presented in Equation (2) and the model is defined as presented on Equation (3). For the signal classification, the output is attributed to the class with the highest *argmax* value.

$$\mathbf{H}\beta = \mathbf{T} \therefore \beta = H^{\dagger}T \tag{1}$$

$$\mathbf{H}^{\dagger} = \mathbf{H^T}(\mathbf{H^T H} + \frac{\mathbf{I}}{\mathbf{C}})^{-\mathbf{1}} \tag{2}$$

$$\beta = \mathbf{H^T}(\mathbf{H^T H} + \frac{\mathbf{I}}{\mathbf{C}})^{-\mathbf{1}}\mathbf{T} \tag{3}$$

D. Classification Ripples and argmax Smoothing

A drawback of *argmax* heuristic is the susceptibility to ripples in *argmax* and the output instability due to the lack of signal representativity. However, this effect can be considered as noisy interference and filtered. This heuristic, considering the sequential classification for ELM was addressed in our previous paper where the ELM using a low-pass filter to mitigate the *argmax* ripples was denominated Smoothed *argmax* ELM (SMT) [28]. In this paper, we replicated the SMT from

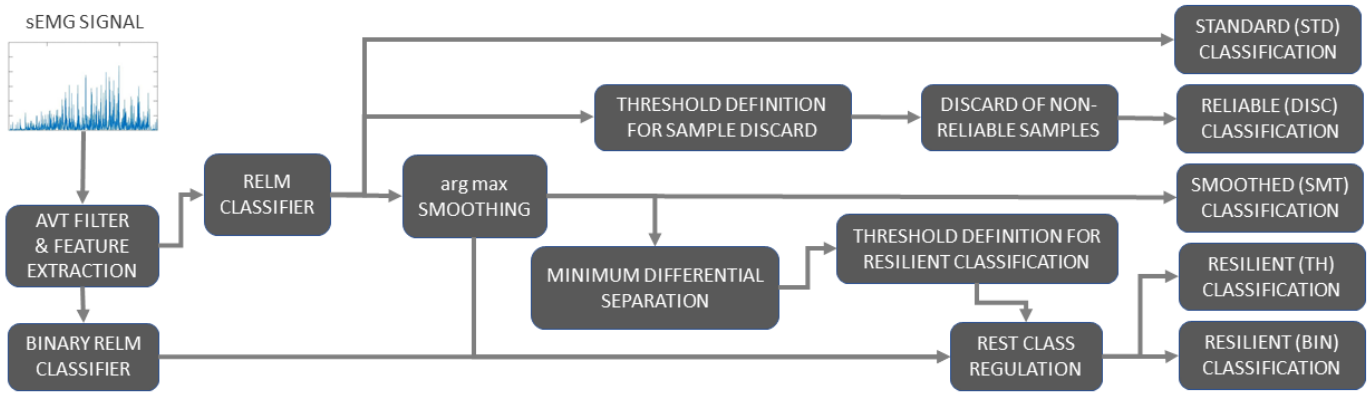

Fig. 1. Overall block diagram for the EMG signal classification with the proposed methods.

our previous work using the AVT as a filter and extended the model to form the resilient classification pipeline. More detailed information about AVT can be found in our previous papers [15], [30].

E. Confidence Score and Reliable Classification

The optimal classification occurs when the *argmax* value for the correct class is as higher and distant as possible from the remaining classes. This distance represents the class separation and the normalized *argmax* absolute value, the probability of each class, or, the *Confidence Score* (*CS*).

Non-reliable classifications are identified by the $CS_{threshold}$ autonomously generated for each subject and classification, as presented by Equation (4). The $\mu_{reliability}$ represents the average *CS* value while $\sigma_{reliability}$ acts as a relaxation factor, considering that correct labelling also occur for *CS* values slightly lower than $\mu_{reliability}$. This forms the *DISCARD* (*DISC*) mode of the classifier, where only reliable samples are considered to generate a more consistent output. Although the potential of sample discard for accuracy boost, the method impairs usability for ignoring the inputs until a reliable classification occurs. To boost accuracy and avoid this delay due to sample discard, the resilient classification models were designed.

$$CS_{threshold} = \mu_{reliability} - \sigma_{reliability} \tag{4}$$

F. Resilient Classification

Two different forms of Resilient classification were tested as an alternative to sample discard (DISC). Since the SMT already handles most of the "in-motion" classification errors, the alternatives were designed to enhance accuracy in movement transitions, where most of the remaining errors reside since the bias leads the classification to rest label instead of an active movement. It was assumed that if an auxiliary method could identify the moment of transition more precisely, the rest class can be ignored from the *argmax* analysis, forcing an active movement label. This process occurs in the REST CLASS REGULATION (RCR) of Figure 1. The resilient forms of the ELM to boost the accuracy in movement transition zones are based on 1) An auxiliary binomial classifier (BIN) and 2) on a new threshold-based method (TH).

For the BIN method, the output of SMT is mapped from multiclass ($C_0 \ldots C_{17}$) to binomial labels ("rest" and "active movement"). In the RCR, a BIN output classified as "rest" results in no interference of the original SMT method. However, an active movement output forces the *argmax* of resting class to be ignored in the label definition. Thus, BIN modulates the use of the rest class *argmax* and cancels the bias that generates the error in movement transitions.

The TH method also aims to provide a binomial signal to modulate the classification from the SMT module. However, instead of an auxiliary classifier, a threshold is generated based on the *Minimum Differential Separation* (*MDS*), as presented by the Equation (5). The *MDS* is the normalized difference between the rest class (C_0) and the closest *argmax* value of the remaining classes ($C_1 \ldots C_{17}$). The *MDS* achieves its maximum value during the rest period and starts to decrease exponentially when an active movement execution starts. Thus, this minimum difference of *argmax* between rest class and active intention serves as an indicator of movement, and it is used to establish the threshold th_{MDS} as presented on Equation (6). Similarly to *CS*, the th_{MDS} is defined by the average of *MDS* (μ_{MDS}) value considering a relaxation factor σ_{MDS}.

Two accuracy rate metrics were used to evaluate the classification: 1) the overall accuracy ($O_{Acc.}$) which is the simple average of all correct samples classified to each volunteer and 2) the weighted accuracy ($W_{Acc.}$), which is the class balanced average accuracy to each subject, as presented in Equation (7).

$$MDS = NORM(min|argmax_{C_0} - argmax_{C_1:C_{17}}|) \tag{5}$$

$$th_{MDS} = \mu_{MDS} - \sigma_{MDS} \tag{6}$$

$$W_{Acc.}\ (\%) = \overline{\sum_{C_0}^{C_{17}} \left(\frac{\text{Correct } class_C \text{ samples}}{\text{Total } class_C \text{ samples}} \right)} \times 100\% \tag{7}$$

III. Experimental Results

Two main tests were performed to evaluate the methods: 1) The accuracy rate according the segmentation for the standard ELM (STD) and 2) The increase of accuracy offered by the reliable and resilient methods compared to STD. Figure 2 presents the overall accuracy rate that emulates a sample-by-sample online classification scenario while Figure 3 presents the weighted accuracy rate, which considers

a class-weighted performance. Despite the overall accuracy useful time-wise indication of error occurrence, it tends to be biased by the predominant rest class that often reaches values close to 100%. Thus, the use of weighted accuracy is also important to assess class-level performance and both metrics were used in a pairwise form to evaluate the system more comprehensively.

A. Segmentation Length × Accuracy Rate

A significant ($p < 0,01$) proportionality between segmentation length and accuracy was confirmed in STD classification. This relation is important for choosing the segmentation considering the system dynamics and proper control since longer segments offer higher accuracy but also insert response delays. While optimal parameters depend on the application, a total delay $< 125ms$ is desirable for proprioceptive control [24].

The accuracy rate was significantly higher ($p < 0,01$) for both metrics with the 400+10ms segmentation. In this configuration, the average overall accuracy of $77,85 \pm 7,14\%$ and $56,74 \pm 21,00\%$, occurred respectively for the NINAPro NAMP and AMP databases while accuracies higher than 95% and 94% were achieved for the sequential (A and B) and random (C and D) assays from the IEE database, respectively. The clear gap in overall accuracy among IEE and NINAPro databases is mainly caused by the rest class bias.

The sequential (A and B) IEE databases reached significantly higher ($p < 0,01$) accuracy rates. IEE A is $\approx 10\%$ to $\approx 20\%$ higher than NAMP considering both accuracy rates. Among IEE databases, despite masked in the overall accuracy, there is a difference of $\approx 10\%$ in the weighted accuracy between sequential and random assays. The AMP database reached lower classification accuracy as expected due to poor model generalization caused by the low representativity of sEMG signals. The reliable and resilient classification methods used the best STD classification scenario, which is 400+10ms.

B. Accuracy Enhancement for Reliable and Resilient Models

Figure 4 presents the auxiliary metrics used by resilient methods and their effect in classification. Although the SMT effectiveness to mitigate classification ripples in the intermediate portion of movements, as presented in Figure 4.b [28], some instability remains in movement transition where the classification tends to be biased towards the rest class. The TH method is dependent on the *MDS* metric which is also plotted in Figure 4.c for a typical classification for Subject 01 of IEE-A database. The nearly perfect movement coverage of the TH is a consequence of the characteristic delay due to the *argmax* filtering that prevents a premature downfall of *MDS* and eliminates the rest class bias at the end of each movement. As presented in Figure 4.d, BIN method is agnostic to the SMT delay and more susceptible to classification ripples between active movements and rest classes. Despite the susceptibility of rest class, Figure 5 presents the consistent accuracy boost to each movement class offered by the BIN method.

Tracking metrics such as the discard rate is necessary to avoid misinterpreted results such as 99% of accuracy while

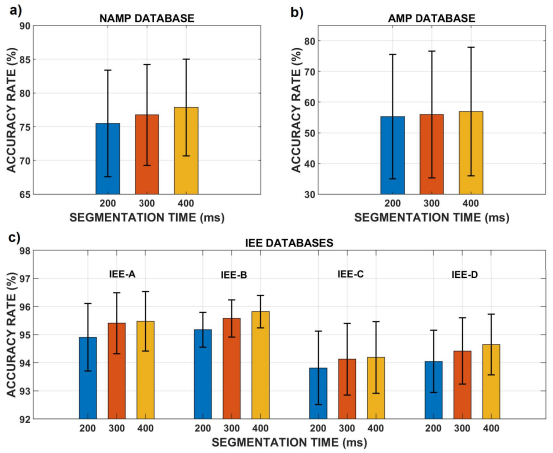

Fig. 2. Effect of segmentation on the overall accuracy.

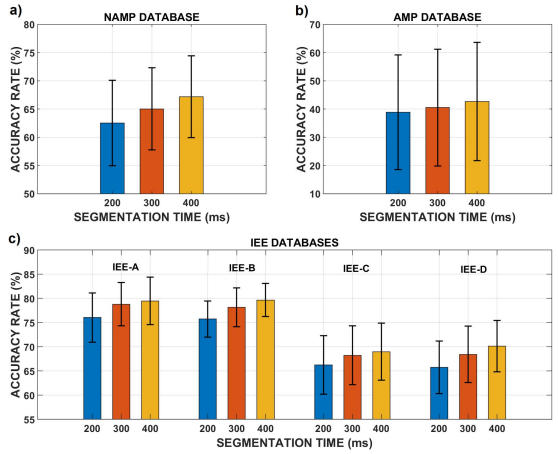

Fig. 3. Effect of segmentation on the weighted accuracy.

discarding 90% of all data. The discard rate varied between $\approx 12\%$ for random assays of the IEE database and $\approx 17\%$ for the NAMP database. The accuracy of the BIN method is also important since an error during the rest class will result in an output error. BIN classifier reached $\approx 78\%$ for AMP data while $\approx 85\%$ and $\approx 97\%$ was obtained for NAMP and IEE databases, respectively. As the DISC method, the TH was activated for every sample where $MDS < th_{MDS}$, affecting between $\approx 15\%$ (IEE-D) and $\approx 19\%$ (NAMP) of samples.

Table II presents the accuracy metrics to all methods where the weighted accuracy presents the potential of the resilient classification. The BIN method reached a higher accuracy for the NINAPro that is formed proportionally with more samples of active movement. In IEE databases, BIN is outperformed by threshold-based methods (DISC and TH). In the NAMP database, TH presents more data coverage and leads to higher weighted accuracy even without data discard. IEE-A and IEE-B are both equivalents considering weighted accuracy (except BIN). A different result was achieved from IEE-C and IEE-D databases, where the random nature of movements affected significantly the *CS*, and DISC performed better.

Figure 5 presents the differential accuracy enhancement at individual movement level compared to STD. The susceptibility of the BIN method to errors in rest class is clear especially for the AMP database where there is a

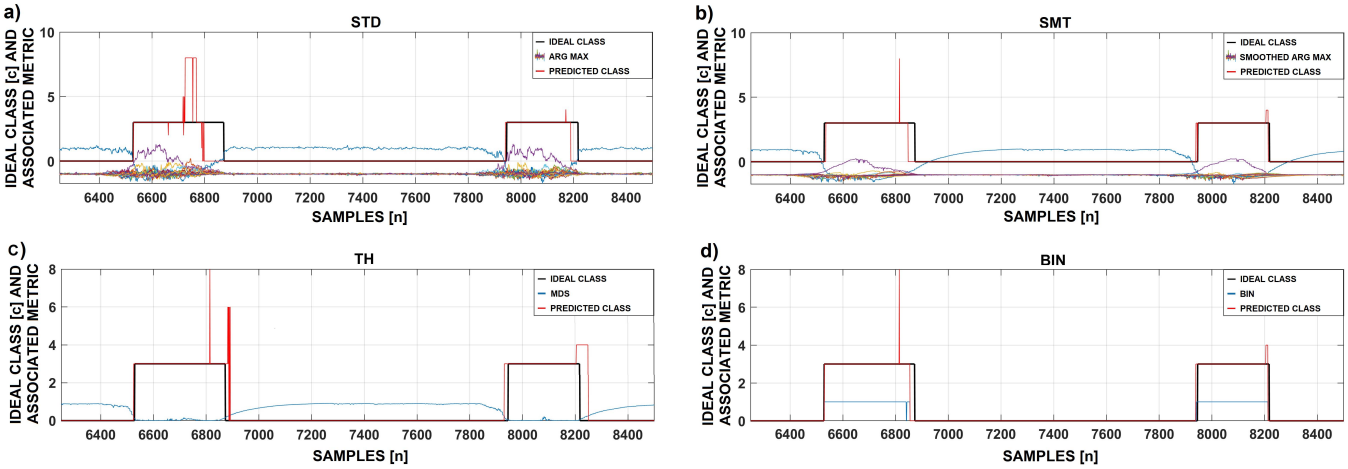

Fig. 4. Auxiliary metrics used in the resilient methods and their effect in the *argmax* signal and classification compared to the standard RELM for Subject 01 of IEE-A database: a) *argmax* derived from the standard (STD) RELM; b) smoothed (SMT) *argmax*; c) Minimum Differential Separation (MDS) signal that supports the resilient classification in the threshold (TH) method and d) auxiliary binomial classification used for the resilient classification in the binomial classifier method (BIN).

TABLE I

AVERAGE OVERALL AND WEIGHTED ACCURACY RATES REACHED FOR ALL THE PROPOSED METHODS TO ALL DATABASES OF THE EXPERIMENT

DATABASE	OVERALL ACCURACY (%)				
	STD	SMT	BIN	TH	DISC
NAMP	$77,9 \pm 7,2$	$81,8 \pm 5,4$	$80,3 \pm 6,5$	$81,7 \pm 5,4$	$87,0 \pm 5,9$
AMP	$56,7 \pm 20,9$	$61,5 \pm 20,8$	$57,9 \pm 24,4$	$61,5 \pm 20,8$	$67,0 \pm 23,6$
IEE A	$95,5 \pm 1,3$	$96,4 \pm 0,9$	$96,5 \pm 1,0$	$93,7 \pm 1,1$	$99,4 \pm 0,4$
IEE B	$95,8 \pm 1,1$	$96,4 \pm 1,0$	$96,6 \pm 1,0$	$93,7 \pm 1,1$	$99,6 \pm 0,3$
IEE C	$94,3 \pm 1,7$	$95,4 \pm 1,7$	$94,8 \pm 1,9$	$94,2 \pm 2,3$	$99,8 \pm 0,4$
IEE D	$94,7 \pm 1,2$	$95,9 \pm 1,1$	$95,5 \pm 1,5$	$95,1 \pm 1,7$	$99,9 \pm 0,1$

DATABASE	WEIGHTED ACCURACY (%)				
	STD	SMT	BIN	TH	DISC
NAMP	$67,3 \pm 7,5$	$75,3 \pm 6,3$	$77,9 \pm 6,2$	$75,8 \pm 6,4$	$73,4 \pm 8,9$
AMP	$42,6 \pm 15,9$	$50,7 \pm 19,0$	$53,4 \pm 18,2$	$51,0 \pm 19,1$	$46,5 \pm 19,3$
IEE A	$79,6 \pm 5,9$	$85,7 \pm 3,9$	$89,0 \pm 3,7$	$91,1 \pm 3,3$	$92,1 \pm 4,4$
IEE B	$79,7 \pm 6,5$	$84,6 \pm 5,0$	$87,9 \pm 4,8$	$90,1 \pm 4,4$	$91,3 \pm 7,0$
IEE C	$69,5 \pm 9,7$	$74,8 \pm 9,1$	$76,0 \pm 9,4$	$77,3 \pm 9,8$	$84,0 \pm 11,5$
IEE D	$70,2 \pm 7,8$	$76,9 \pm 6,7$	$78,6 \pm 6,8$	$80,7 \pm 6,5$	$87,8 \pm 7,0$

decrease of $\approx$ 20% for movement 1 (rest) due to the low data representativeness. The AMP database also presented the lower accuracy rate for the binomial classifier ($\approx$ 78%). For the NAMP database, the error induced at the rest class is half as much ($\approx$ 10%) since the signals are more descriptive. However, for AMP and NAMP the error in the rest class is compensated by a considerable increase of accuracy for all movements leading the BIN method to be the most accurate for the NINAPro database. Table II presents the results for all methods. While BIN presented an increase of $\approx$ 10% for NAMP and AMP databases, SMT and TH presented similar but slightly lower results for all movements. However, the methods still offer an increase of $\approx$ 8% of weighted accuracy compared to STD for NINAPro databases. The limitations of SMT and TH rely on incapacity to cover the whole movement and the delay at the end of each movement, as presented in Figure 4. For NINAPro, the resilient methods offered higher accuracy when compared to the DISC method, even without sample discarding.

For the IEE database, the overall performance of the classifiers is maintained. However, due to the bias of the longer rest period, the effect of TH is attenuated, leading to higher accuracy rates compared to the BIN. This indicates the potentiality of the TH method when the filter delay is mitigated. Overall, all resilient methods present a significant accuracy increase from $\approx$ 8% (IEE-C) to $\approx$ 12% (IEE-A) when compared to the baseline classifier without any sample discard. These results are only outperformed by the DISC method, which matches the TH performance for IEE-A and IEE-B databases and reached $\approx$ 15% and $>$ 17% of accuracy increase for the IEE-C and IEE-D databases, respectively.

IV. DISCUSSION

A. Accuracy Rate $\times$ Segmentation Length and Baseline ELM

Despite the intrinsic proportion between segmentation length and sEMG signal representativity, the sample buffering can cause delays that prevent proprioceptive control. In this

paper, the optimal scenario was achieved using 400+10ms segments that served as the baseline for all the post-processing methods proposed. The use of 400ms provides representative information about sEMG signal while 10ms enable agile steps to maintain an appropriate processing time (ideally, <125ms) for real-time control. The rest class is quite different from other classes since it is characterized by minimal or no signal amplitude and tends to reach higher accuracy even in poorly generalized models. As a consequence, a consistent difference between overall and weighted accuracy is clear and despite it generates a misleading idea about the classifier performance, this difference is often omitted in the literature.

The sequential IEE databases reached the highest weighted accuracy while the random assays still perform better than NINAPro. The IEE-A was designed according to the NINAPro and follows the same experimental protocol. The other subsets of the IEE database were designed to evaluate the consistency of classification on a more extensive set of repetitions and also check potential bias induced by the sequential order of the movements, which, led the assays (IEE-C and IEE-D) to lower accuracy rates. The volunteers from AMP database also reached lower accuracy, translating the difficulty to obtain accurate models for amputee volunteers due to the lack of representativeness in their signals. Although some AMP volunteers are capable of reaching satisfactory results, most of them lack specific musculature for prosthetic control. Even though this can be compensated by subject training and alternative electrode positioning, this was not explored in the AMP database. Due to the electrode positioning, two volunteers were incapable of using two sEMG channels placed on the forearm and these channels are the main responsible for the finger movements that alone represent a third of the movements in the database. Those two volunteers present a low accuracy rate even for the rest class. This lack of representation remains an open problem in the literature, affecting the model generation and leading to inaccurate results. Overall, considering the NINAPro database, the baseline (STD) ELM classifier tends to match or outperform recent papers in literature.

B. Accuracy Enhancement for Reliable and Resilient Models

Despite the STD accuracy, errors in movement transition and classification ripples were still frequent in the output. Most of the classification ripples tend to be mitigated by SMT that also inserts a lag for the transition sections due to the filter characteristic phase delay. Since the SMT is the base of the resilient methods, where all the *argmax* from STD was smoothed by AVT. There is a direct relationship between the filter used for the SMT and the responsiveness of the TH method, where a filter with a longer phase delay will cause proportional latency after each movement, which is also presented in Figure 5.c. Although the delay leads to a decrease of the rest class accuracy by prolonging the movement deactivation, experimental tests are still needed to quantify this characteristic and different filters can be used to reduce this setback. Despite its consistent accuracy improvement to all databases, the TH was outperformed by the BIN and DISC to NINAPro and IEE databases, respectively.

The results of Table II, present the potential of the resilient classification only for the weighted accuracy since overall accuracy is heavily biased by the rest class. In the weighted analysis, a consistent advantage of the BIN method is presented for the NINAPro database while IEE database results vary according to the dataset. The threshold definition for the DISC method considers all classes, and it is influenced by the resting class higher average reliability value. Thus, the threshold of DISC is increased, and the method is prone to over-rejection, leading to higher accuracy but precluding the proprioceptive control. The over-rejecting effect represents more a limitation of the DISC than an advantage of the resilient methods since DISC relies on a class balance to calculate its reliability threshold autonomously. Nevertheless, the resilient methods (BIN or TH) were able to approximate or even surpass the DISC accuracy in the remaining databases, even without sample discard. The Table II presents the summary for the resilent methods application and results.

C. Comparison With Related Works

Related studies from Atzori *et al.* [29] with traditional approaches and NINAPro NAMP database used methods such as k-NN, SVM, Random Forests, and LDA along with variations of features where their best results reached a 75,3% of overall accuracy rate. Zhai *et al.* [22] reached 75,7% of the overall accuracy rate for the NINAPro base using a spectrogram and dimensionality reduction with PCA as pre-processing alternatives. In their paper, Shi *et al.* [34] used different comparisons of features based on entropy for the classification of six different wrist and hand movements to derive a feature called CREn with results varying from 94% to 97% depending on segmentation used (from 200ms to 1s) on a self-generated database. In our study, the STD version of our method reached $\approx$ 78% of the overall accuracy rate for the NAMP database. Moreover, the accuracy was increased to $\approx$ 82% using the TH method and $\approx$ 87% using DISC, presenting a blatant enhancement compared to Atzori *et al.* [29] and Zhai *et al.* [22]. Moreover, the use of 400 + 10ms for segmentation is a configuration more feasible than 1s considering proprioceptive control. Related to the IEE database, overall accuracies higher than $\approx$ 90% of weighted accuracy rate were obtained for the sequential assays (IEE-A and IEE-B) using the TH and DISC methods. Moreover, consistent enhancements were also observed in random assays. For the overall accuracy, all methods (including the STD), reached at least $\approx$ 95% of accuracy rate. Additionally, the DISC method reached results of $\approx$ 100% for all the assays of the IEE database.

Compared to recent approaches, Zhai *et al.* [22] used CNN as the base for a regenerative model that enhances $\approx$ 10% of the overall accuracy compared to their standard classifier (SVM). However, their method relies on a considerable amount of processing for real-time classification and the definition of many hyperparameters. Those are two metrics in which ELM takes great advantage not only compared with Deep Learning but also with more traditional classifiers. Moreover, their accuracy increase is given compared to overall classification while our method increase provides a more spe-

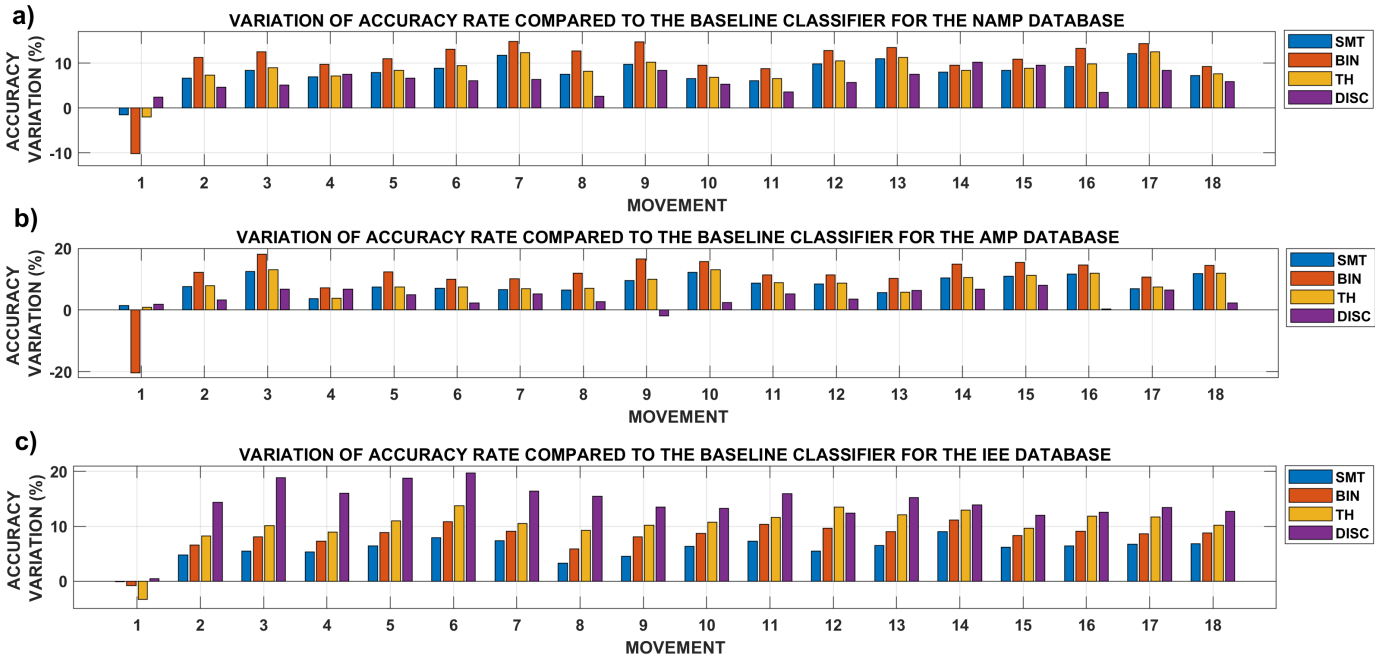

Fig. 5. Resilient methods and DISC individual movement accuracy in comparison to the baseline RELM. a) represents the effect in NAMP database while b) presents the effects in AMP database and c) the overall rates for the IEE database.

TABLE II
SUMMARY OF THE RESILIENT CLASSIFICATION PIPELINE COMPARED TO THE DISC METHOD

METHOD	DESCRIPTION	PROS	CONS
SMT	*argmax* filtering to eliminate classification ripples.	Low complexity (relies solely on arg max filtering) and is easily adaptable to different filters.	Efficiency is linked to filter behavior and filters with higher attenuation tend to increase the phase delay (responsivity).
BIN	Auxiliar classifier to modulate the rest class usage.	Intuitive modulation of the rest class usage and significant improvement for amputee subjects.	Errors during rest in binomial classifier lead to output errors and the use of two classifiers requires the double of memory.
TH	Uses the MDS to modulate the rest class usage.	High coverage of active movements (efficient reduction of rest influence during movements).	Response delay related to the SMT filter and the threshold definition sensitivity to imbalanced datasets.
DISC	Identify and discard non-reliable samples autonomously.	Substantial improvement of accuracy by considering only the reliable outputs.	An excessive threshold level induces over-rejection of data and precludes the proper proprioceptive control.

cific solution and increases considerably the weight accuracy that represents a general enhancement of all classes.

The direct comparison with papers that use alternative metrics, or hybrid models to generate more consistent classification is often impractical given the different task-related metrics. However, similar implementation to our DISC method can be found on papers such as [8], [26], [27], all of them focusing on increasing the system robustness through sample discard that although efficient to increase accuracy, tend to preclude proprioceptive control. Robertson *et al.* [11] explored a confidence score based on Fitts Law instead of an intrinsic classifier metric such as the *argmax*. The authors also evaluate different levels of threshold and their influence on the accuracy rate and highlight the benefit of including a reliability metric to reduce systemic (classifier) errors and how an excessively restricting value can preclude usability due to the excess of sample discard. That is precisely where the resilient classification techniques present advantages compared to the discard methods since they are capable to approximate (IEE database) or increase (NINAPro databases) the accuracy rate even without any sample discard.

V. CONCLUSION

Among the outcomes, we found that the use of the RELM and simple time-domain features combined with the DES and a proper segmentation (400+10ms) leads to higher or compatible results concerning the literature. The resilient classifiers proved to regulate the rest class bias in movement transition and increase the accuracy, which is a major drawback in literature. Results indicated that the resilient classifiers are suitable for robust sEMG classification and accuracy enhancement that in some cases outperforms our previous reliable classification while avoiding sample discard. Future investigations will focus on the different filters for *argmax* smoothing, thresholds refinement, and confidence scores to

build regenerative models that can benefit long-term use of the interface.

Conflict of Interest Statement

The authors do not have relations with other people/ organizations that could bias their work.

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